

BECKLEY, OH, USA

WMO: 724295

Lat: 39.840N

Lon: 83.840W

Elev: 1051

StdP: 14.15

Time Zone: -5.00 (NAE)

Period: 07-19

WBAN: 63888

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.9	6.3	-6.4	4.1	2.4	-1.0	5.4	7.2	27.2	29.7	24.7	29.2	6.9	280	0.497

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.3	90.3	75.5	87.9	74.3	85.8	73.1	79.2	86.3	77.3	84.7	75.7	82.6	8.5	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.3	147.8	83.9	75.2	137.6	81.4	73.6	129.9	79.6	43.5	84.9	41.5	84.6	39.8	82.7	
(p)															
86.2															

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.9	20.3	18.1	DB	-8.3	93.4	7.0	3.5	-13.3	95.9	-17.4	97.9	-21.3	99.9	-26.4	102.4	(4)
(5)			WB	-8.3	81.2	6.9	2.4	-13.2	82.9	-17.2	84.3	-21.1	85.6	-26.1	87.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.4	26.9	29.6	41.2	52.3	63.8	71.4	73.6	72.2	66.5	54.8	41.6	33.3
	DBStd	18.99	12.33	12.63	11.67	9.53	8.40	5.68	4.95	4.88	7.27	9.50	9.87	9.97
	HDD50	2581	721	580	324	85	4	0	0	0	1	57	286	523
	HDD65	5546	1182	992	738	393	129	13	2	5	69	338	704	981
	CDD50	3455	4	8	53	153	432	642	733	687	495	207	34	7
	CDD65	944	0	0	2	11	92	205	269	228	113	23	1	0
	CDH74	8238	0	0	23	121	853	1727	2320	1859	1099	233	3	0
	CDH80	2607	0	0	3	16	230	579	795	553	372	59	0	0
Wind	WSAvg	8.1	10.2	10.1	9.9	9.9	8.2	7.0	5.4	4.8	5.6	7.7	8.8	9.6
Precipitation	PrecAvg	41.0	3.1	2.2	3.0	3.9	4.7	4.4	4.8	3.2	3.1	2.9	2.7	3.0
	PrecMax	52.6	9.4	4.6	6.9	7.5	8.0	7.8	8.4	6.4	6.8	5.2	4.5	7.8
	PrecMin	29.6	1.3	0.7	0.9	1.1	1.4	2.2	1.7	0.6	0.8	0.7	0.7	1.0
	PrecStd	6.1	1.8	1.0	1.3	1.5	2.0	1.4	1.6	1.4	1.7	1.5	1.1	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.3	66.6	77.3	82.3	88.4	92.1	94.1	91.6	91.9	86.4	72.2	64.0
		MCWB	57.9	58.8	63.8	63.1	71.0	74.2	80.5	77.0	73.3	71.7	58.4	60.8
	2%	DB	56.1	59.1	70.6	77.8	85.3	89.0	90.0	88.0	87.9	80.2	66.8	58.4
		MCWB	53.2	54.6	60.4	62.3	68.9	73.8	77.4	75.8	73.1	66.5	56.8	55.2
	5%	DB	50.4	54.9	64.2	73.2	82.9	86.3	87.3	85.5	84.5	75.5	61.8	54.0
		MCWB	47.9	50.9	56.3	59.3	68.2	72.4	75.6	74.2	71.5	64.0	54.0	51.7
	10%	DB	44.7	48.2	59.9	68.8	79.7	83.4	84.8	83.1	80.8	70.8	57.1	48.9
		MCWB	41.3	44.1	54.1	58.0	67.2	70.9	74.6	72.6	68.5	62.2	50.5	45.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.4	60.3	64.6	67.1	75.0	79.0	82.6	80.7	79.8	73.2	63.6	60.9
		MCDB	61.1	65.7	73.6	75.4	84.4	85.9	89.3	87.3	85.7	83.4	70.7	63.3
	2%	WB	53.8	55.7	61.3	64.1	72.4	76.7	80.1	78.0	75.5	69.9	59.0	56.1
		MCDB	56.0	58.0	68.2	72.2	81.3	85.3	87.5	84.7	84.4	75.8	63.9	57.7
	5%	WB	48.3	51.2	58.4	62.0	70.6	75.2	78.0	76.3	73.4	66.7	55.0	51.9
		MCDB	50.0	54.7	62.8	69.2	78.9	83.3	85.3	82.9	80.6	72.3	60.8	53.9
	10%	WB	41.4	44.2	54.7	59.8	68.7	73.4	76.2	74.6	71.2	63.6	51.2	46.1
		MCDB	43.5	48.0	59.4	67.2	75.9	79.9	82.4	80.6	77.4	69.6	56.3	48.7
Mean Daily Temperature Range	5% DB	MDBR	15.2	16.4	18.9	21.6	21.1	19.8	19.3	20.4	22.6	21.0	18.2	13.9
		MCDBR	18.5	21.9	24.9	27.2	24.4	23.3	21.7	22.1	26.4	26.2	24.0	20.0
		MCWBR	16.9	18.4	16.2	15.2	11.3	10.8	10.7	11.1	12.7	14.0	15.9	18.2
	5% WB	MCDBR	18.1	21.1	20.9	21.6	20.6	19.5	19.7	19.1	21.9	21.7	20.6	19.4
MCWBR		17.4	18.7	15.1	14.7	11.1	10.2	11.7	10.2	11.6	13.0	15.7	18.8	
Clear-Sky Solar Irradiance	taub	0.300	0.324	0.345	0.388	0.434	0.443	0.446	0.442	0.389	0.354	0.322	0.303	
	taud	2.443	2.347	2.368	2.265	2.174	2.200	2.196	2.190	2.319	2.412	2.460	2.454	
	Ebn at Noon	275	283	288	281	269	266	264	261	270	268	265	265	
	Edh at Noon	25	32	35	42	47	45	45	44	36	30	25	23	
All-Sky Solar Radiation	RadAvg	537	813	1127	1520	1745	1924	1879	1731	1413	990	667	454	
	RadStd	49	96	98	125	144	128	109	87	129	90	82	46	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (32 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+202	+113

Nomenclature: See separate page